THE ARTIN-SCHREIER TYPE POLYNOMIAL P(Z) OBTAINED FROM A PRIMITIVE PTH ROOT OF UNITY HAS COEFFICIENTS IN THE IDEAL GENERATED BY W = 1 - ZETA AND SATISFIES AN ADDITION FORMULA

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The author writes 'The second part follows from a direct computation we omit'; that sentence is kept verbatim and marks the one step left to the reader. The proof also contains the author's typo 'the proof if the first part is done', which is reproduced unchanged rather than silently corrected.

Allowed standard prerequisites (editorial).

- The ring of integers Z[zeta] of a cyclotomic field and its unique prime above p. the reduction at proof lines 34992-34995 and the prime-ideal statement at proof lines 35003-35005. Supplies the unique prime (w) above p and hence the relation p = u w^{p-1} that drives the computation of the coefficients a_i.
- Divisibility of cyclotomic polynomials and the identity p = prod_j (1 - zeta^j). the factorization and its evaluation at t = 1 at proof lines 34996-35002. Is the arithmetic input that identifies the valuation of p at w as p - 1.

Lemma 0.2. *Let A be a ring which contains a primitive pth root of unity ζ. Set $w = 1 - \zeta$. Then*

$$P(z) = \frac{(1+wz)^p - 1}{w^p} = z^p - z + \sum\nolimits_{0<i<p} a_i z^i$$

is an element of $A[z]$ and in fact $a_i \in (w)$. Moreover, we have

$$P(z_1 + z_2 + wz_1z_2) = P(z_1) + P(z_2) + w^p P(z_1)P(z_2)$$

in the polynomial ring $A[z_1, z_2]$.

Proof. It suffices to prove this when

$$A = \mathbf{Z}[\zeta] = \mathbf{Z}[x]/(x^{p-1} + \ldots + x + 1)$$

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is the ring of integers of the cyclotomic field. The polynomial identity $t^p - 1 = (t-1)(t-\zeta)\ldots(t-\zeta^{p-1})$ (which is proved by looking at the roots on both sides) shows that $t^{p-1} + \ldots + t + 1 = (t-\zeta)\ldots(t-\zeta^{p-1})$. Substituting $t = 1$ we obtain $p = (1-\zeta)(1-\zeta^2)\ldots(1-\zeta^{p-1})$. The maximal ideal $(p, w) = (w)$ is the unique prime ideal of A lying over p (as fields of characteristic p do not have nontrivial pth roots of 1). It follows that $p = uw^{p-1}$ for some unit u. This implies that

$$a_i = \frac{1}{p}\binom{p}{i} uw^{i-1}$$

for $p > i > 1$ and $-1 + a_1 = pw/w^p = u$. Since $P(-1) = 0$ we see that $0 = (-1)^p - u$ modulo (w). Hence $a_1 \in (w)$ and the proof if the first part is done. The second part follows from a direct computation we omit. $\square$